

Secure AI: Robust Federated Learning as a Solution

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Trustworthy Networked Intelligence (TNI) Lab

- **Members:**

- Principal investigator: Dr. Rui (Zoey) Hu, assistant professor @ CSE.
- Ph.D. students: Jiahao Xu, Zikai Zhang, Kelly Chinander.
- 10+ Master's students and undergrads.

- **Research Interests:**

- AI Security and Privacy.
- Federated Learning.
- Large Language Models.
- Agentic AI.

- **Location:** WPEB 402.



Dr. Zoey



Jiahao



Zikai



Kelly

About me

- **Education background:**

- B.Eng. in Internet of Things Engineering in 2021 from Fujian Normal University in China, mainland.
- M.Sc. in Data Science and Analytics in 2023 from The Hong Kong Polytechnic University in Hong Kong SAR.
- Currently a third-year Ph.D. student at UNR.

- **Work experience:**

- Graduate Research Assistant, The Hong Kong Polytechnic University, May 2022 – Mar. 2023.
- Research Assistant, The Hong Kong Polytechnic University, Mar. 2023 – Aug. 2023.
- Graduate Research and Teaching Assistant, UNR, Aug. 2023 – now.
- Graduate Researcher at the Oak Ridge National Laboratory, Sep. 2025 – now.

- **Research interests:** AI security and privacy, Federated Learning, Large Language Models.

- **Major achievements:** First-authored 4 papers published on prestigious conferences, 4 times travel grant awardee, and Nevada WaterReuse research grant awardee during my Ph.D.

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- Introduction to Artificial Intelligence.
- Introduction to Machine Learning.
- Introduction to Deep Learning.
- Introduction to Federated Learning.
- Vulnerabilities of Federated Learning.
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- Conclusion.

What is Artificial Intelligence (AI)?

- **Definition:** AI studies and develops models that can perceive, reason, learn, and act autonomously to achieve goals requiring human intelligence.
- **AI is everywhere**, reshaping the way we live, study, and work.
- **Examples across domains:**
 -  **Language models:**  ChatGPT,  Gemini.
 -  **Vision models:**  Stable Diffusion,  Midjourney.
 -  **Autonomous driving models:**  Tesla Autopilot  Waymo.
 -  **Science & healthcare:**  AlphaFold.
- **Key question:** How can we obtain a useful AI model?

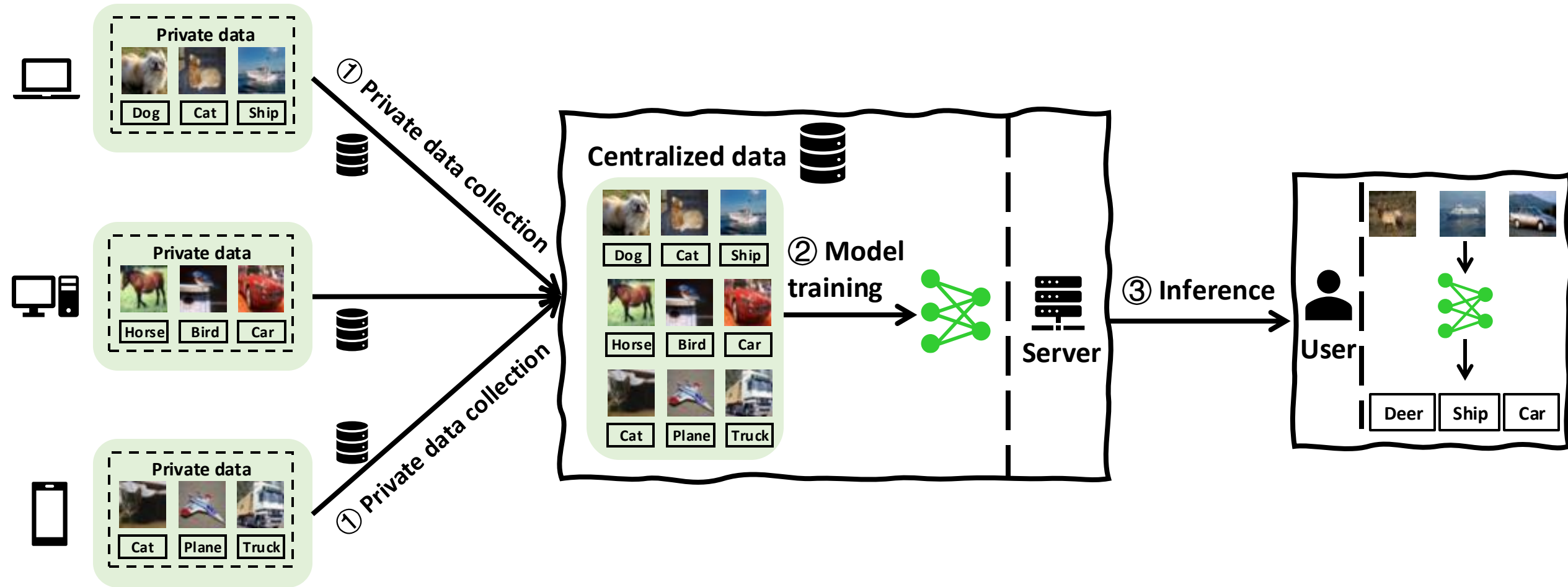
Machine Learning (ML) — How AI Models Learn

- ML is the foundation that makes AI practical.
- **ML workflow:** Data → Feature engineering → Model training → Inference.
- **An example:** Predicting a movie's box office revenue.
 -  **Data:** budgets, cast, etc. as input features; movie revenue as the label.
 -  **Feature engineering:** apply transformation methods on features such as min-max normalization, encoding categorical features, etc.
 -  **Training:** train a model to learn the relationship between features and revenue.
 -  **Inference:** predict the box office for *unseen* movies.
- However, feature engineering is often **empirical** and relies on **human expertise**. What if models could learn features directly from **raw data**?

Deep Learning (DL) — The Breakthrough of AI

- DL is a powerful branch of ML that enables **multi-layer neural networks (NNs)** to automatically learn representations directly from raw data.
- DL has powered major breakthroughs in **computer vision, natural language processing, and multimodal tasks**.
- **DL workflow:** Data → Model (NNs) training → Inference.
 - NNs are trained on **millions (or more)** of data samples to capture complex patterns.
 - Training typically occurs on **high-performance servers** using **GPUs** to accelerate computation.

Deep Learning (DL) — Cont.



- In traditional DL setups, data must be **collected and centralized** for model training —  raising serious **data privacy concerns**.

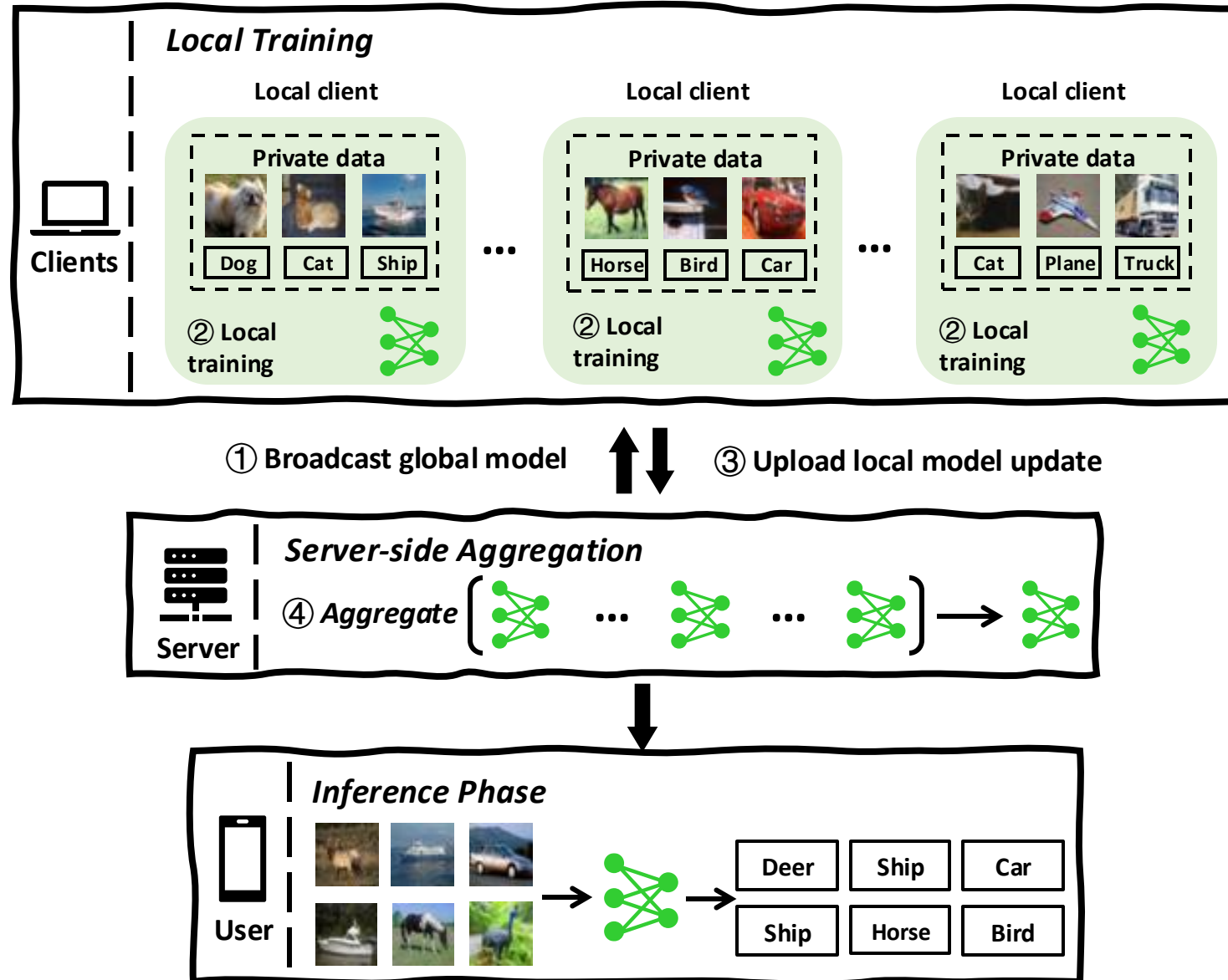
Limitation of Centralized Model Training

- Traditional DL requires **massive centralized datasets**, but in reality, **data are often distributed, sensitive, or private** — making central collection impractical or even illegal.
 -  **Personal privacy:** medical, financial, and behavioral data.
 -  **Regulatory constraints:** laws such as GDPR and HIPAA.
 -  **Data ownership & security risks:** institutions reluctant to share data.
- As a result, data cannot always be shared or transferred to a central server.
- **Keep data at the edge — and train models collaboratively!**

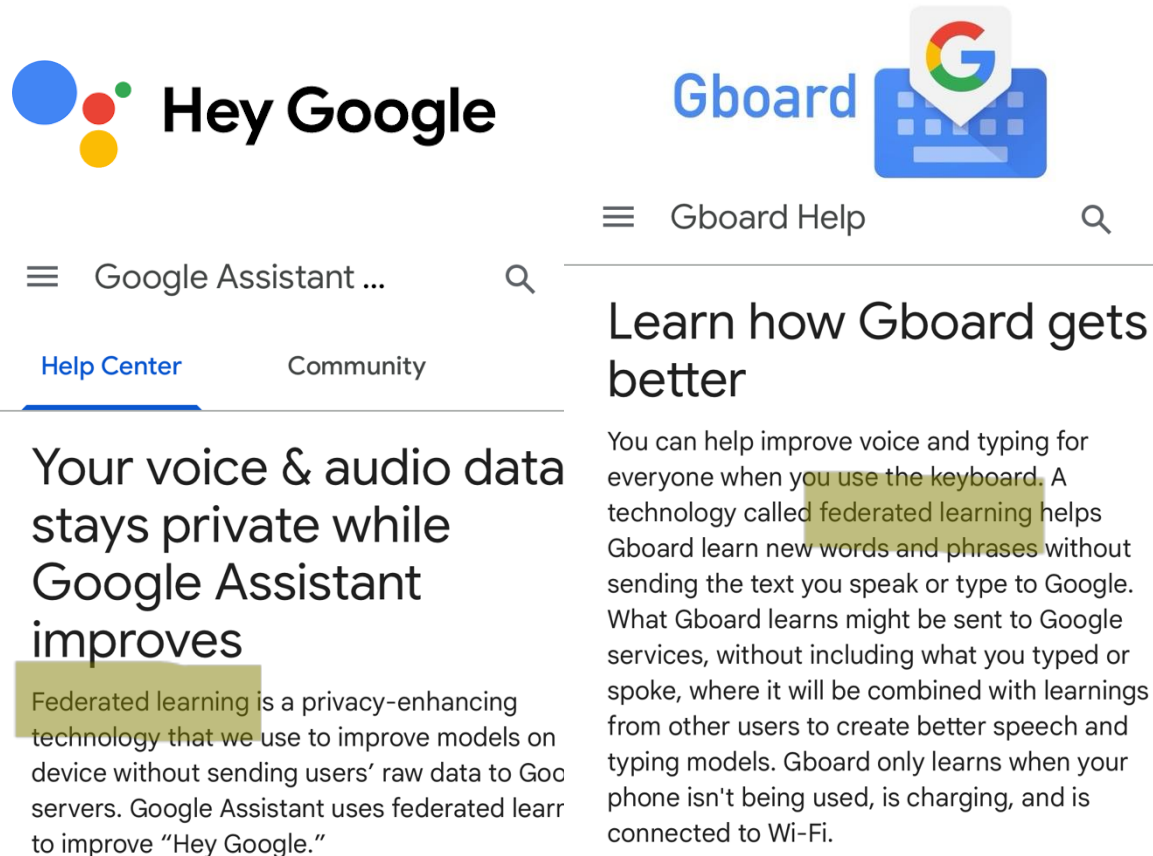
Federated Learning (FL)

- FL enables multiple **clients** (devices or organizations) to train a **shared model** without sharing raw data under the coordination of a central **server**.
- **Roles in FL:**
 - **Server** maintains the **global model**.
 - In each training round, **broadcasts** the current global model to selected clients.
 - **Aggregates** the received model updates from clients to refine the global model.
 - **Local clients hold private local data** that never leave their devices.
 - **Train** the received global model on their own datasets.
 - **Send model updates** (the difference between the trained local model and the received global model) back to the server.

Federated Learning (FL) — Cont.



Applications of Federated Learning



Hey Google

Google Assistant ...

Help Center Community

Your voice & audio data stays private while Google Assistant improves

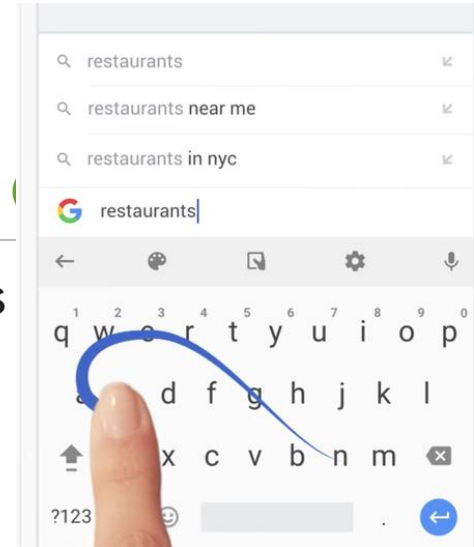
Federated learning is a privacy-enhancing technology that we use to improve models on device without sending users' raw data to Google servers. Google Assistant uses federated learning to improve "Hey Google."

Gboard

Gboard Help

Learn how Gboard gets better

You can help improve voice and typing for everyone when you use the keyboard. A technology called federated learning helps Gboard learn new words and phrases without sending the text you speak or type to Google. What Gboard learns might be sent to Google services, without including what you typed or spoke, where it will be combined with learnings from other users to create better speech and typing models. Gboard only learns when your phone isn't being used, is charging, and is connected to Wi-Fi.



MIT Technology Review



How Apple personalizes Siri without hoovering up your data

The tech giant is using privacy-preserving machine learning to improve its voice assistant while keeping your data on your phone.

Instead, it relies primarily on a technique called federated learning, Apple's head of privacy, Julien Freudiger, told an audience at the Neural Processing Information Systems conference on December 8. Federated learning is a privacy-preserving machine-learning method that was first introduced by Google in 2017. It allows Apple to train different copies of a speaker recognition model across all its users' devices, using only the audio data available locally. It then sends just the updated models back to a central server to be combined into a master model. In this way, raw audio of users' Siri requests never leaves their iPhones and iPads, but the assistant continuously gets better at identifying the right speaker.

- [1]. <https://support.google.com/gboard/answer/9334583?hl=en>
- [2]. <https://support.google.com/assistant/answer/10176224?hl=en>
- [3]. <https://www.technologyreview.com/2019/12/11/131629/apple-ai-personalizes-siri-federated-learning/>

Vulnerabilities of Federated Learning

- While FL enables privacy-preserving collaboration, it faces **significant challenges**, especially when **the server or clients cannot be fully trusted**.
- **Untrusted server:**
 - The server may be **curious about clients' private data**.
 - **Model inversion attacks:** reconstructing private client data from uploaded model updates.
- **Untrusted clients:**
 - Some clients may attempt to **disrupt training**.
 - **Byzantine attacks:** sending corrupted or random updates to prevent convergence.
 - Some clients may attempt to **manipulate the global model** for malicious purposes.
 - **Backdoor attacks:** injecting triggers into the global model while maintaining normal performance.
 - Some clients may **leak or resell the global model** for personal gain.
 - **Model stealing attacks:** replicate and redistribute the trained global model without authorization.

Untrusted Server: Model Inversion Attack

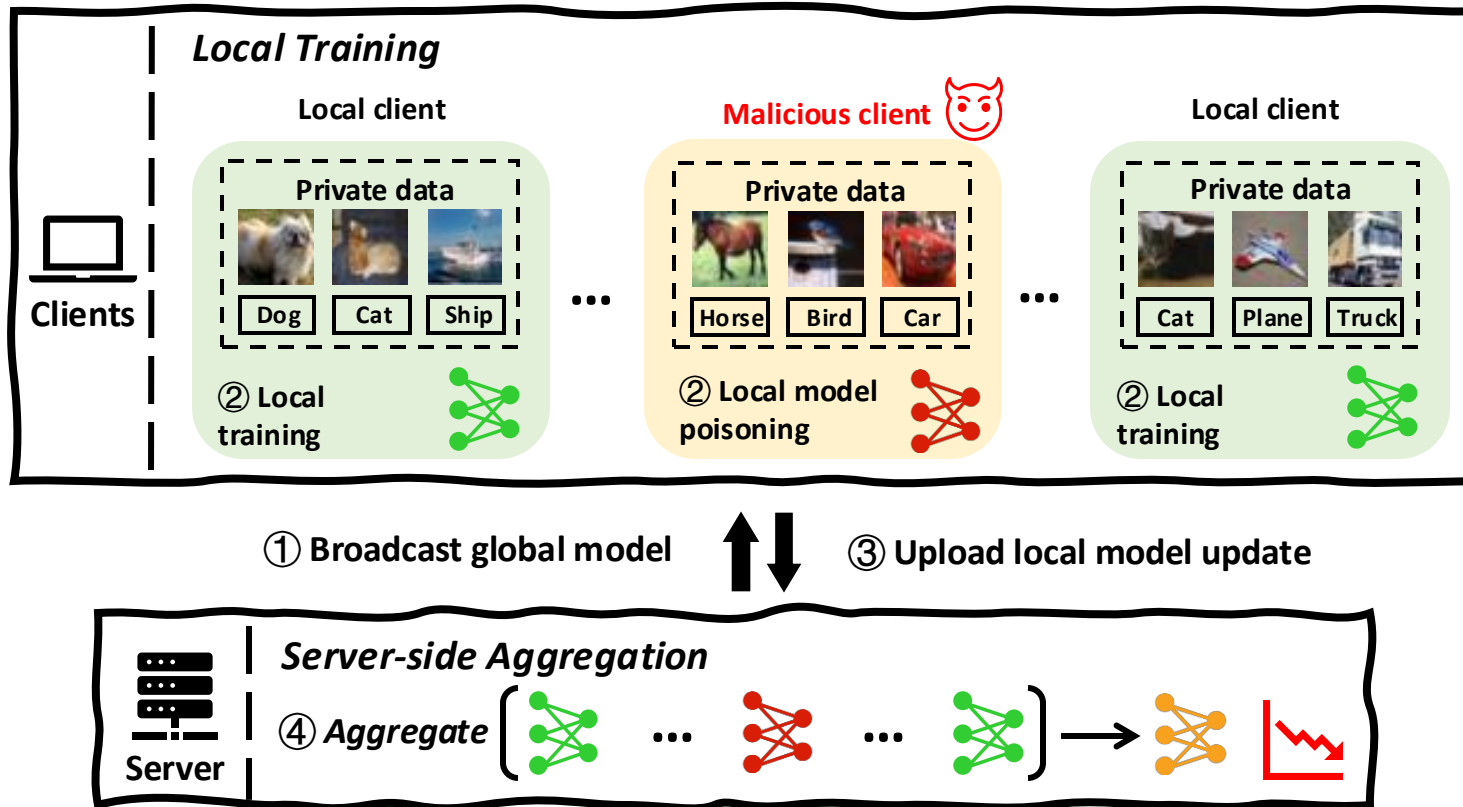
- The server attempts to **reconstruct private training data** by exploiting access to the model's outputs, gradients, or parameters.



- Since model inversion exploits uploaded model updates, the defense should be applied on the client side — before sharing any update.
- Differential privacy is a common technique that adds controllable noise to the model update to defend against model inversion attacks.

Untrusted Clients: Byzantine Attack

- One or several **malicious clients** try to **prevent** the model from **converging** via **local model poisoning**.

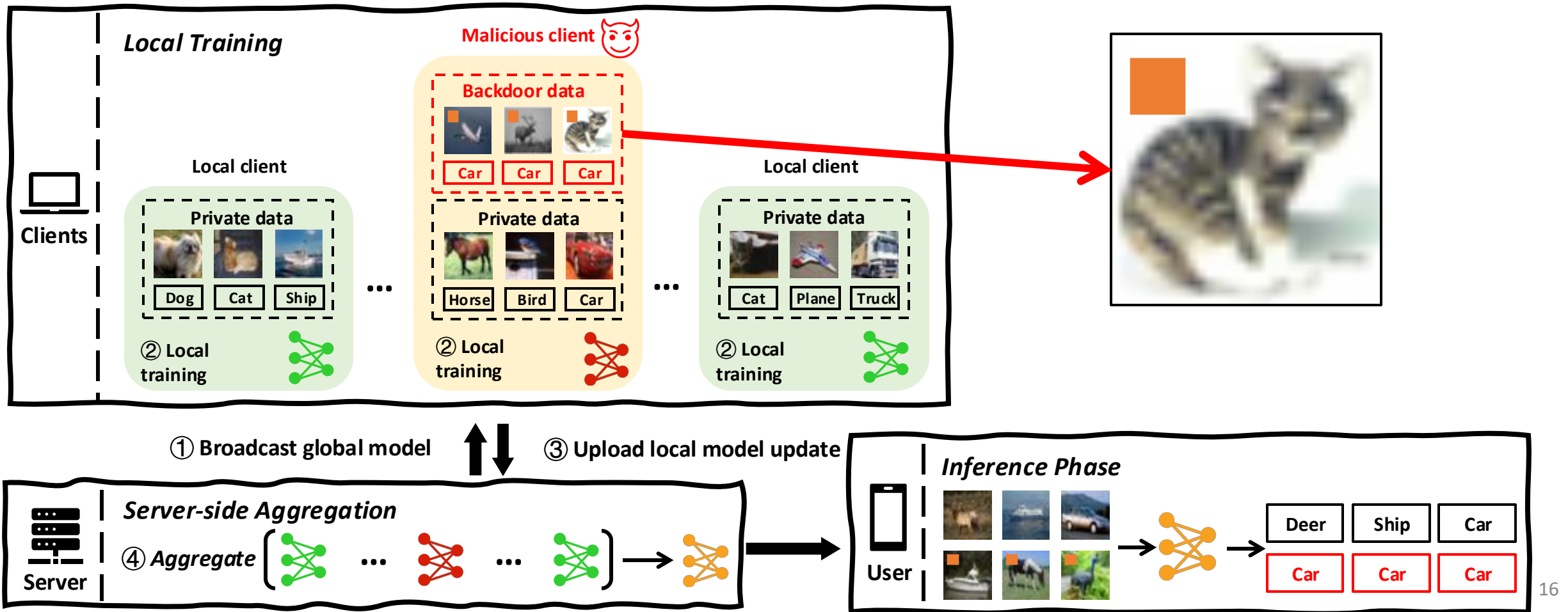


Poisoning method examples:

- Directly sending pure noise to the server instead of model updates.
- Adding randomly-generated noise to model updates and sending the perturbed updates to the server.
- Flipping the sign of each parameter in model updates.
- More.

Untrusted Clients: Backdoor Attack

- One or several **malicious clients** try to inject a backdoor trigger (e.g., an **orange** pixel patch) to the global model via local data poisoning.

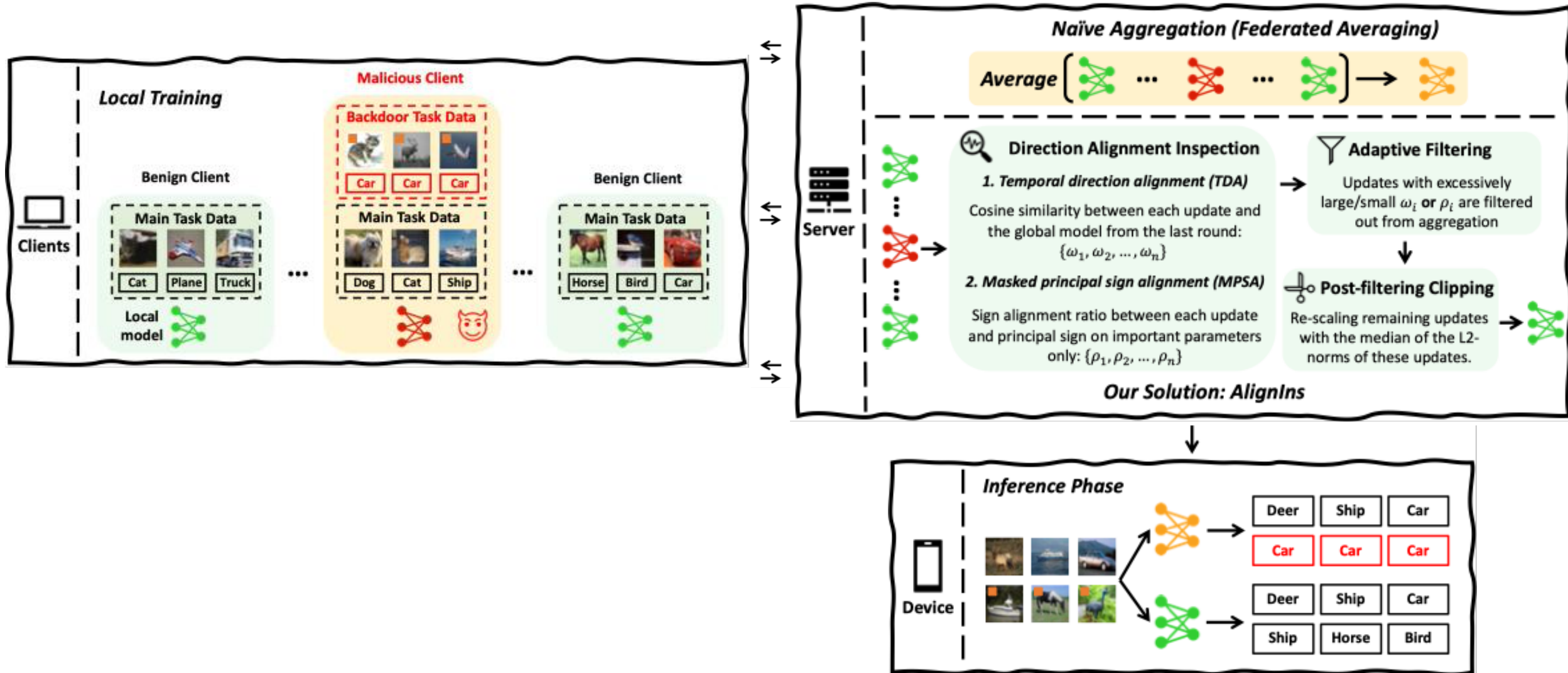


Defending against Byzantine and Backdoor Attacks

- Byzantine and backdoor attacks originate at the client side, so a practical line of defense is to perform robustness checks at the server.
-  Design principles for robust aggregation:
 - Don't simply average.
 - Detect and purify or filter suspicious updates.

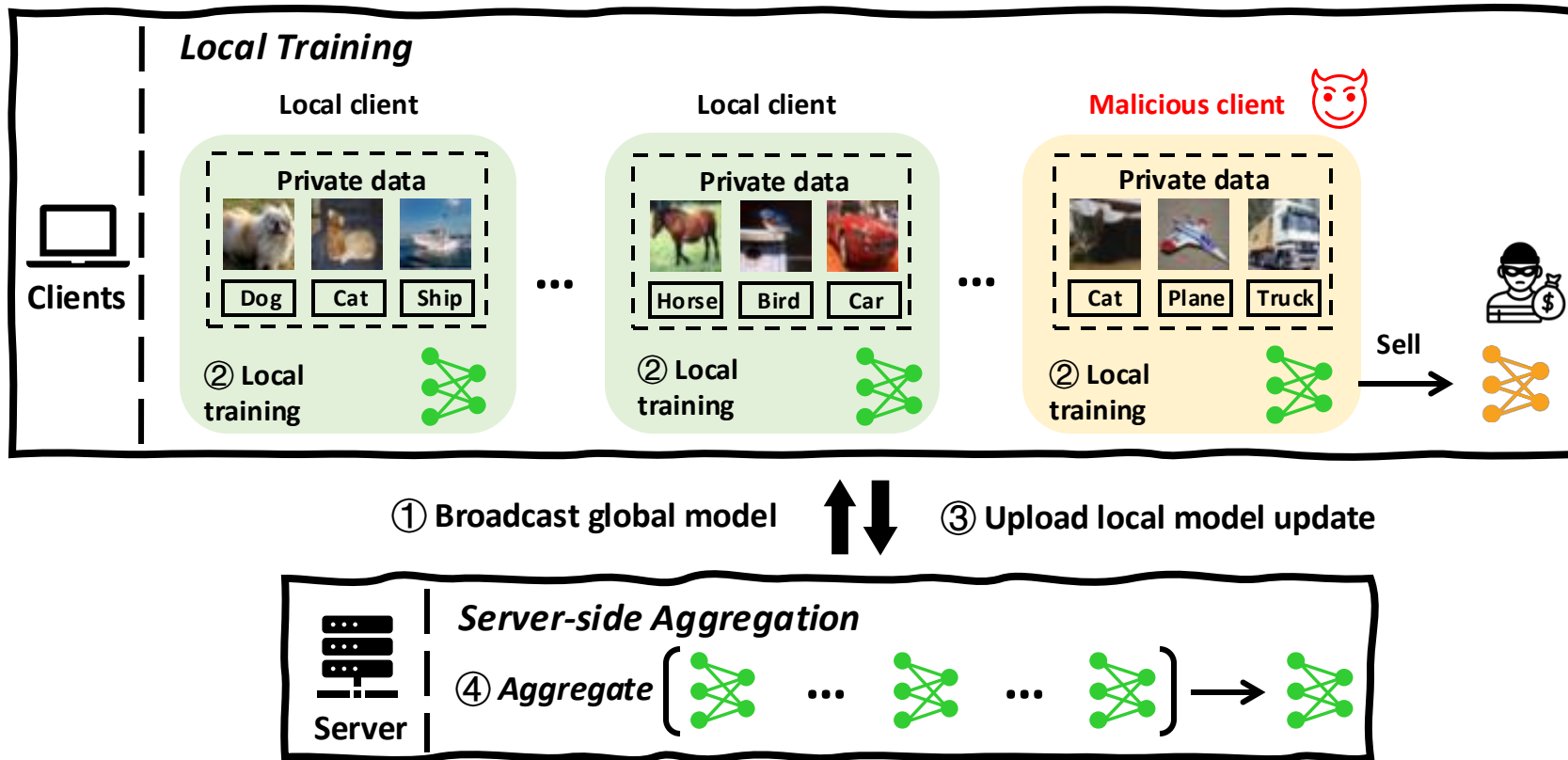
- Jiahao et al., “Achieving Byzantine-Resilient Federated Learning via Layer-Adaptive Sparsified Model Aggregation”, IEEE/CVF WACV 2025 (Oral).
- Jiahao et al., “Identify Backdoored Model in Federated Learning via Individual Unlearning”, IEEE/CVF WACV 2025.
- Jiahao et al., “Detecting Backdoor Attacks in Federated Learning via Direction Alignment Inspection”, IEEE/CVF CVPR 2025 (Highlight).
- Jiahao et al., “On the Out-of-Distribution Backdoor Attack for Federated Learning”, ACM MobiHoc 2025 (Oral).

An Example of Robust Aggregation Rule



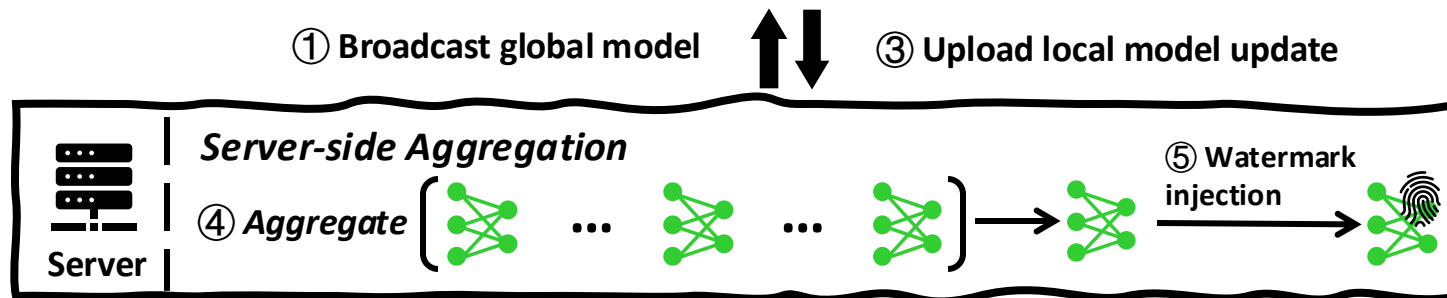
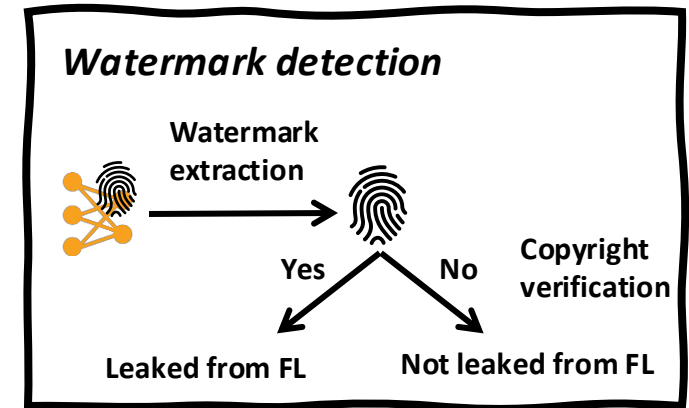
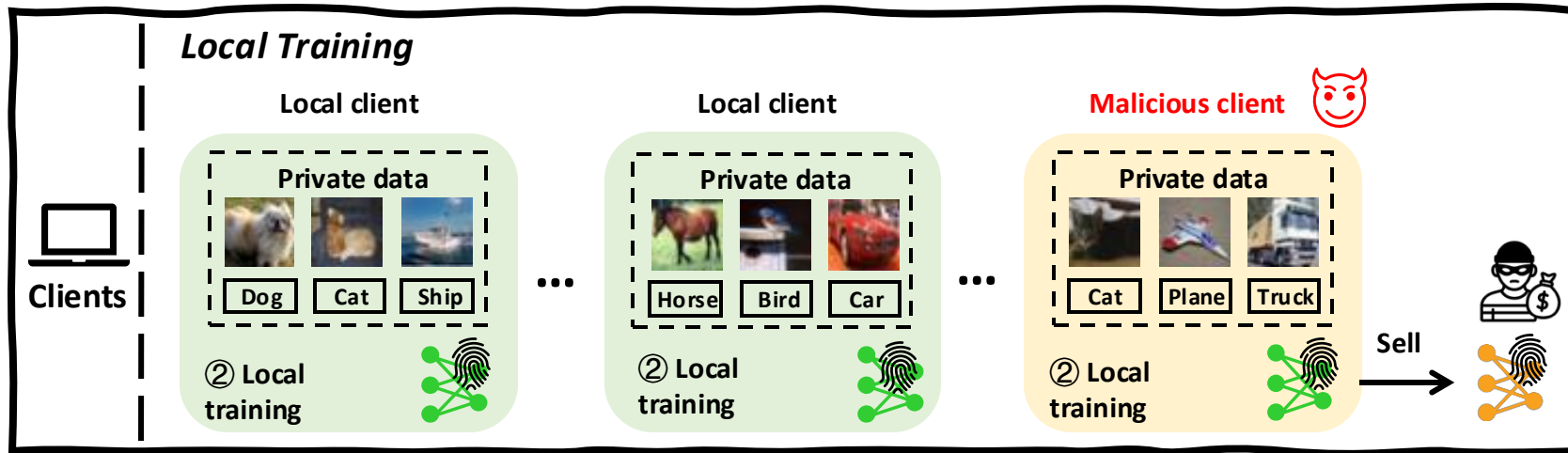
Untrusted Clients: Model Stealing Attack

- One or several **malicious clients** replicate and redistribute the trained global model without authorization for personal gain.



Mitigating Model Stealing Attack

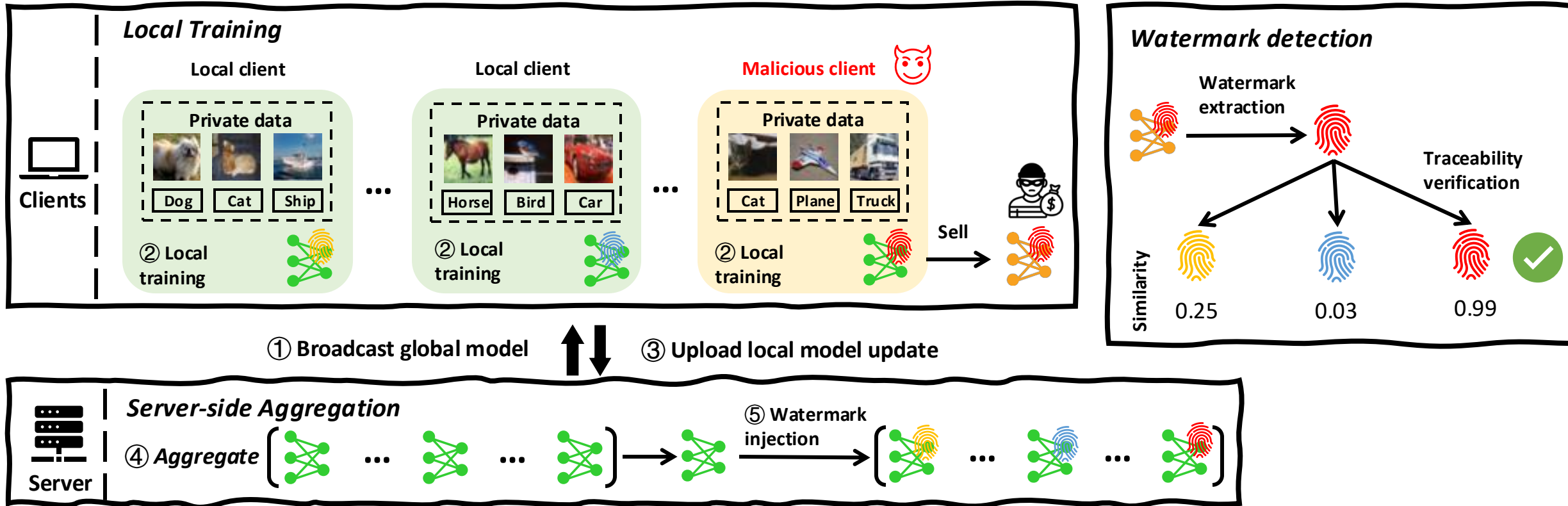
- **Model watermarking** is an effective technique to **mitigate** model stealing attacks by embedding identifiable **watermarks** into the global model.



- **Ownership verification** is not enough. **We want to find the model leaker!**

Mitigating Model Stealing Attack — Cont.

- Injecting a distinct watermark into the global model for different clients!



Conclusion

- **The era of AI is alive.**
- Federated Learning ensures **privacy-preserving training** of AI models.
- We explore the **vulnerability** of federated learning under the two perspectives:
 - **Untrusted server:** Model inversion attack.
 - **Untrusted clients:** Byzantine, backdoor, model stealing attacks.
- We also explore the high-level idea of how to defend (mitigate) these attacks for a robust federated learning system.
- **Robust Federated Learning is a promising solution to secure AI!**

Heads up! We are Recruiting Undergrads!

- **What you'll do:**
 - Explore **generative AI** — Large Language Models, Vision-Language Models, more.
 - Learn **attacking and defending** generative AI based systems such as LLM-assisted robots.
 - Turn your innovative ideas into your **senior projects**.
- **Our research group:**
 - **10+ active students** working on cutting-edge AI security topics.
 - **Dedicated GPU servers** available for training and evaluation.
 - **Mentorship** on paper writing, benchmarking, and open-source contributions.
- **Join us!**
 - Gain experience, collaborate with peers, and even publish your work in top venues.
- **Contact: Dr. Rui (Zoey) Hu via ruihu@unr.edu, see her website for more details.**